Permutation Tests for Vectors, Networks and Tables

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Module 3: Resample

Computational Methods in Statistics and Data Science (DATASCI 406)

Review: the permutation recipe

From deck 4:

1. H_0: some rearrangement leaves the distribution of the data unchanged.
2. **Condition on** what H_0 says nothing about.
3. **Rearrange** the rest. Every rearrangement is equally likely under H_0.
4. **Compute** T on each rearrangement and count.

Deck 4 used single numbers. This deck: vectors, networks and tables.

Two null hypotheses for multivariate data

Equal distributions. $X_1, \ldots, X_m \overset{\text{iid}}{\sim} F$ and $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} G$, each in $\mathbb{R}^p$.

$$H_0 : F = G.$$

Shuffle the group labels, as in deck 4.

Independence. Pairs (X_i, Y_i), $i = 1, \ldots, n$, iid.

$$H_0 : F_{XY}(x, y) = F_X(x) F_Y(y).$$

Shuffle the Y_i against the X_i.

Vectors have no single ordering, so the statistics are built from **pairs of observations**.

Statistics Built from Pairs

Pairwise statistics

$$T = \sum_{i=1}^{n} \sum_{j=1}^{n} A_{ij} B_{ij}, \qquad A, B \text{ are } n \times n.$$

- A_{ij}: distance between i and j, or a network tie.
- B_{ij}: same label, or a distance in a second variable.

This is the **Mantel statistic**. Every test in this deck has this form.

Relabeling observations in a pairwise statistic

A permutation π of the observations moves row i **and** column i of B together:

$$T^* = \sum_{i,j} A_{ij} B_{\pi(i)\pi(j)}.$$

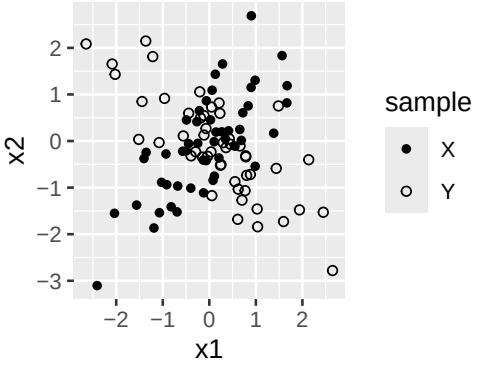
```
mantel <- function(A, B) sum(A * B)
mantel_null <- function(A, B, reps) {
  n <- nrow(A)
  replicate(reps, {
    p <- sample(n)
    mantel(A, B[p, p])
  })
}
```

Two Samples of Vectors

Margins and joint distributions

$F = G$ means equal **joint** distributions, not equal margins.

Example: $N(0, 1)$ margins, correlation 0.7 for X and -0.7 for Y.



Kolmogorov-Smirnov tests on each coordinate

```
c(coordinate_1 = ks.test(X[, 1], Y[, 1])$p.value,
  coordinate_2 = ks.test(X[, 2], Y[, 2])$p.value)
```

```
## coordinate_1 coordinate_2
##       0.2719       0.9667
```

Every margin is $N(0, 1)$, so coordinate-wise tests have no power here for any m.

Energy distance

Euclidean distance between $x, y \in \mathbb{R}^d$:

$$\|x - y\| = \sqrt{\sum_{k=1}^{d} (x_k - y_k)^2}.$$

For $X, X' \overset{\text{iid}}{\sim} F$ and $Y, Y' \overset{\text{iid}}{\sim} G$, independent, with finite means,

$$\mathcal{E}(F, G) = 2\mathrm{E}\left(\|X - Y\|\right) - \mathrm{E}\left(\|X - X'\|\right) - \mathrm{E}\left(\|Y - Y'\|\right).$$

- $\mathcal{E}(F, G) \geq 0$, and $\mathcal{E}(F, G) = 0$ only when $F = G$.
- Mixed samples: distances between groups match distances within, so $\mathcal{E} \approx 0$.

Energy distance from a sample

Replace each expectation with an average over the pooled distance matrix $D_{ij} = \|Z_i - Z_j\|$:

$$\hat{\mathcal{E}} = \frac{2}{mn} \sum_{X,Y} D_{ij} - \frac{1}{m^2} \sum_{X,X} D_{ij} - \frac{1}{n^2} \sum_{Y,Y} D_{ij}.$$

Reject for large values. Scaling by $mn/(m+n)$ does not change the p-value.

Energy statistic in R

Compute `dist()` once. A shuffle changes only which entries are averaged.

```
energy <- function(D, g) {
  2 * mean(D[g, !g]) - mean(D[g, g]) - mean(D[!g, !g])
}
D_sim <- as.matrix(dist(sim))
e_obs <- energy(D_sim, g_sim)
e_null <- replicate(B, energy(D_sim, sample(g_sim)))
c(observed = e_obs, p = pval(e_null, e_obs))
```

```
## observed        p
##   0.1812   0.0220
```

Rejects where the coordinate-wise KS tests did not.

Energy statistic as a pairwise statistic

With weights $L_{ij} = 1/m^2$ within X, $1/n^2$ within Y, and $-1/(mn)$ between,

$$\hat{\mathcal{E}} = -\sum_{i,j} D_{ij} L_{ij}.$$

So $A = D$ and $B = -L$.

```
nx <- sum(g_sim); ny <- sum(!g_sim)
L <- outer(g_sim, g_sim, function(a, b)
  ifelse(a & b, 1 / nx^2, ifelse(!a & !b, 1 / ny^2, -1 / (nx * ny))))
c(energy = energy(D_sim, g_sim), mantel = mantel(D_sim, -L))
```

```
## energy mantel
## 0.1812 0.1812
```

Nearest-neighbor statistic

$N(i)$ is the closest other point to i in the pooled sample.

$$T_{NN} = \sum_{i=1}^{m+n} \mathbf{1}\{g_i = g_{N(i)}\}.$$

- Under $F \neq G$, points cluster with their own sample and T_{NN} is large.
- **One-sided**: a small T_{NN} (more mixed than chance) is not evidence against H_0.
- Variant: count same-label neighbors among the r nearest.

As a pairwise statistic: $A_{ij} = 1$ if $j = N(i)$, and $B_{ij} = 1$ if $g_i = g_j$.

Nearest-neighbor test in R

Neighbors do not depend on the labels, so find them once.

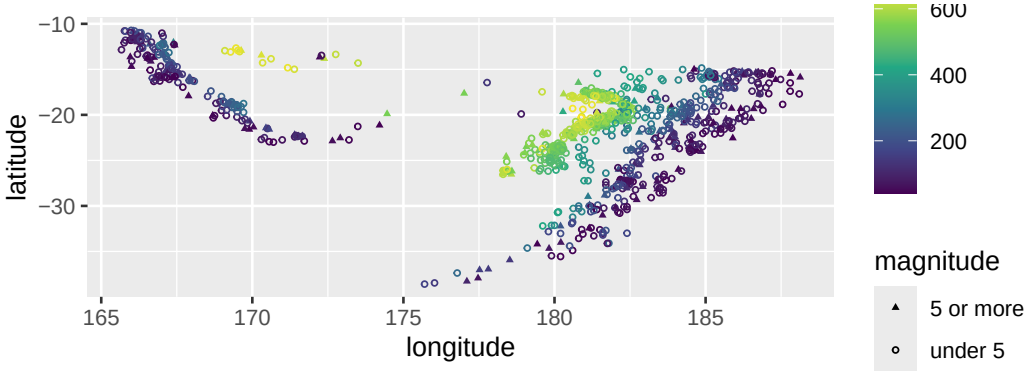
```
nearest <- function(D) { diag(D) <- Inf; apply(D, 1, which.min) }
t_nn <- function(nn, g) sum(g == g[nn])
nn_sim <- nearest(D_sim)
nn_obs <- t_nn(nn_sim, g_sim)
nn_null <- replicate(B, t_nn(nn_sim, sample(g_sim)))
c(observed = nn_obs, null_mean = mean(nn_null), p = pval(nn_null, nn_obs))
```

```
##  observed null_mean         p
##    69.000    49.574     0.002
```

Example: earthquakes near Fiji

quakes (shipped with R): 1000 earthquakes near Fiji since 1964, magnitude 4 or more.

H_0: locations (latitude, longitude, depth) of magnitude 5+ earthquakes have the same distribution as the rest.



Choosing units for the distance

Latitude and longitude are in degrees, depth in kilometers.

```
sapply(q[, c("lat", "long", "depth")], sd)
```

```
##     lat    long   depth
##   5.029   6.069 215.535
```

- Raw scale: the distance is almost entirely depth.
- `scale()`: each coordinate in SDs, weighted equally.

Neither is correct. The units set each coordinate's weight in the distance.

Energy and nearest-neighbor tests on the earthquake data

```
D_q <- as.matrix(dist(scale(q[, c("lat", "long", "depth")])))
e_obs <- energy(D_q, g_q)
e_null <- replicate(B, energy(D_q, sample(g_q)))
nn_q <- nearest(D_q)
nn_obs <- t_nn(nn_q, g_q)
nn_null <- replicate(B, t_nn(nn_q, sample(g_q)))
c(energy = pval(e_null, e_obs), nearest_neighbor = pval(nn_null, nn_obs))
```

```
##           energy nearest_neighbor
##            0.001            0.139
```

Same shuffles. Energy uses every distance, nearest neighbor one per point, and it has less power here.

Networks as adjacency matrices

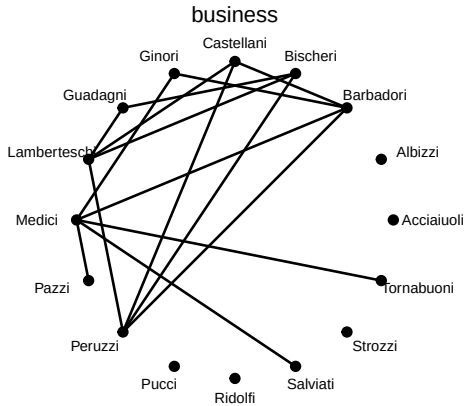
Adjacency matrix: $A_{ij} = 1$ if nodes i and j are linked. The nearest-neighbor graph was one.

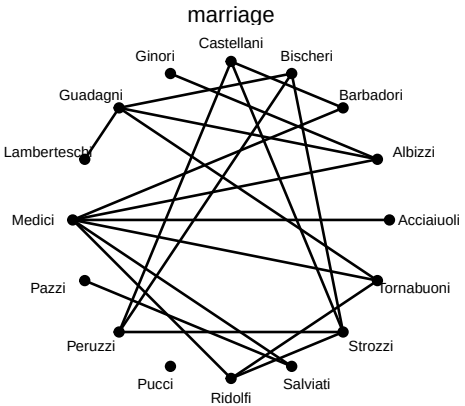
- Are two networks on the same nodes related? A, B both adjacency matrices.
- Is a node attribute related to the network? B_{ij} compares i and j.

QAP (quadratic assignment procedure): under H_0, relabel the nodes of one matrix.

Example: Florentine families

Marriage and business ties among 16 leading Florentine families around 1430 (Padgett).





Testing association between two networks

H_0: the marriage and business networks are unrelated.

T: pairs tied by both, $\frac{1}{2}\sum_{i,j} M_{ij}B_{ij}$.

```
t_obs <- mantel(marriage, business) / 2
t_null <- mantel_null(marriage, business, B) / 2
c(observed = t_obs, null_mean = mean(t_null), p = pval(t_null, t_obs))
```

```
##  observed null_mean         p
##     8.000     2.530     0.001
```

8 shared pairs, against 2.5 expected.

Testing a node attribute against the marriage network

H_0: wealth in 1427 is unrelated to marriage. Relabel wealth among families.

T: the average wealth gap across marriages. $B_{ij} = |w_i - w_j|$.

```
W <- abs(outer(fam$wealth, fam$wealth, "-"))
gap <- function(W) mantel(marriage, W) / sum(marriage)
g_obs <- gap(W)
g_null <- replicate(B, { p <- sample(k); gap(W[p, p]) })
c(observed = g_obs, null_mean = mean(g_null),
  p_two_sided = 2 * min(pval(g_null, g_obs), pval(-g_null, -g_obs)))
```

```
##    observed   null_mean p_two_sided
##      57.300      38.451       0.038
```

Marriages joined families further apart in wealth than chance.

Summary: two samples of vectors

- Equal margins do not imply $F = G$.
- Energy uses every pairwise distance and is 0 only when $F = G$.
- Nearest neighbor counts same-label neighbors and is one-sided.
- Shuffles do not change distances, so compute them once.
- The units set each coordinate's weight in the distance.

Independence

Testing independence by shuffling the pairing

H_0: X and Y are independent, with (X_i, Y_i) iid.

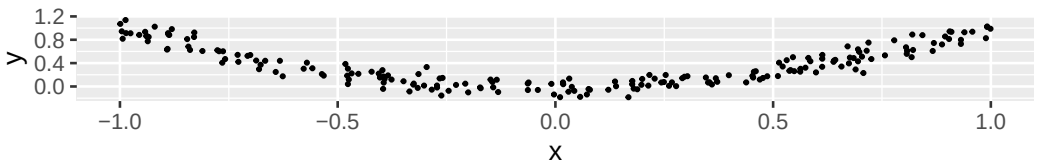
- **Condition on**: the X values and the Y values.
- **Rearrange**: the pairing. Every matching of Ys to Xs is equally likely, $n!$ in all.
- T: any measure of dependence.

The Florentine tests were of this kind.

Pearson correlation and nonlinear dependence

```
x <- runif(200, -1, 1)
y <- x^2 + rnorm(200, sd = 0.1)
r_null <- replicate(B, cor(x, sample(y)))
c(r = cor(x, y), p = pval(abs(r_null), abs(cor(x, y))))
```

```
##         r         p
## -0.02613  0.71900
```



Y is nearly a function of X, but r is near 0 because the relationship is not linear.

Distance covariance

Distance matrices $a_{ij} = |X_i - X_j|$ and $b_{ij} = |Y_i - Y_j|$, each **double-centered**:

$$\tilde{a}_{ij} = a_{ij} - \bar{a}_{i\cdot} - \bar{a}_{\cdot j} + \bar{a}_{\cdot\cdot}$$

Squared sample distance covariance, a pairwise statistic:

$$\mathrm{dCov}_n^2(X, Y) = \frac{1}{n^2} \sum_{i,j} \tilde{a}_{ij} \tilde{b}_{ij}.$$

- The population version is 0 **only** under independence (finite means).
- For vectors, use $\|X_i - X_j\|$.

Distance correlation

Scale by the distance variances, $\mathrm{dVar}_n^2(X) = \mathrm{dCov}_n^2(X, X)$:

$$\mathrm{dCor}_n^2(X, Y) = \frac{\mathrm{dCov}_n^2(X, Y)}{\sqrt{\mathrm{dVar}_n^2(X)\,\mathrm{dVar}_n^2(Y)}}.$$

- $0 \leq \mathrm{dCor} \leq 1$, and $\mathrm{dCor} = 0$ only under independence.
- The sample dCor is positive even under independence. Compare it with its permutation null.

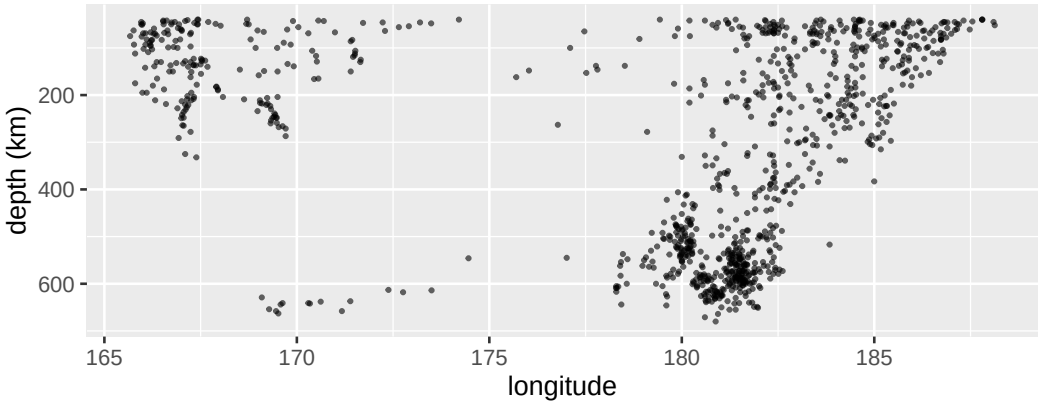
Distance correlation in R

Centering $b[\pi, \pi]$ gives $\tilde{b}[\pi, \pi]$, so center once, then permute.

```
center <- function(D) D - outer(rowMeans(D), colMeans(D), "+") + mean(D)
dcor <- function(A, B) sqrt(mean(A * B) / sqrt(mean(A^2) * mean(B^2)))
A <- center(as.matrix(dist(x)))
Bc <- center(as.matrix(dist(y)))
d_obs <- dcor(A, Bc)
d_null <- replicate(B, { p <- sample(length(y)); dcor(A, Bc[p, p]) })
c(dcor = d_obs, max_null = max(d_null), p = pval(d_null, d_obs))
```

```
##     dcor max_null        p
##   0.4623   0.2393   0.0010
```

Example: earthquake depth and longitude



West of 175°: mostly shallow. East: depth decreases with longitude, along the plate sinking under Tonga.

Pearson and distance correlation on the earthquake data

```
A_q <- center(as.matrix(dist(q$long)))
B_q <- center(as.matrix(dist(q$depth)))
d_obs <- dcor(A_q, B_q)
d_null <- replicate(B, { p <- sample(nrow(q)); dcor(A_q, B_q[p, p]) })
east <- q$long > 175
c(r = cor(q$long, q$depth), r_east = cor(q$long[east], q$depth[east]),
  dcor = d_obs, max_null = max(d_null))
```

```
##         r    r_east      dcor max_null
##    0.1444  -0.6557    0.4091    0.1080
```

With 1000 points both reject, but r is small because the two regions slope in opposite directions.

Tables

Two binary variables as a permutation test

$W_i, Z_i \in \{0, 1\}$ for N individuals give a 2×2 table.

- Shuffle Z against W. The margins stay fixed, so one cell sets the table.
- $T = A_{11}$, the number with $W = 1$ and $Z = 1$.

This is **Fisher's exact test**. Shuffling gives the hypergeometric null without the formula.

Example: graduate admissions at Berkeley, 1973

`UCBAdmissions` (shipped with R): applications to the six largest departments.

```
ucb <- as.data.frame(UCBAdmissions) %>% uncount(Freq)
ucb_admit <- ucb$Admit == "Admitted"
ucb_female <- ucb$Gender == "Female"
dept <- ucb$Dept
```

Pooled over departments, 30% of 1835 women and 45% of 2691 men were admitted.

H_0: admission is independent of gender.

Pooled test of admission and gender

$T = A_{11}$, women admitted. One-sided: were women admitted less often?

```
a_obs <- sum(ucb_admit & ucb_female)
a_null <- replicate(B, sum(ucb_admit & sample(ucb_female)))
c(observed = a_obs, null_mean = mean(a_null),
  p_lower = pval(-a_null, -a_obs))
```

```
##  observed null_mean   p_lower
##   557.000   711.669     0.001
```

557 women were admitted, against 712 expected under independence.

Admission rates by department

Dept	applicants	pct_female	admit_female	admit_male
A	933	12	82	62
B	585	4	68	63
C	918	65	34	37
D	792	47	35	33
E	584	67	24	28
F	714	48	7	6

Women applied mostly to selective departments. Within departments, they were admitted at about men's rate or higher.

Shuffling within strata

H_0: within each department, admission is independent of gender.

- **Condition on**: each applicant's department, and each department's totals.
- **Rearrange**: gender, **within** department only.
- T: $\sum_k A_{11k}$, women admitted, summed over departments.

Stratified test for the Berkeley data

`shuffle_within()` shuffles z within each level of s. Rows keep their order.

```
shuffle_within <- function(z, s)
  tibble(z, s) %>% group_by(s) %>% mutate(z = sample(z)) %>% pull(z)
s_null <- replicate(B, sum(ucb_admit & shuffle_within(ucb_female, dept)))
c(observed = a_obs, null_mean = mean(s_null),
  p_two_sided = 2 * min(pval(s_null, a_obs), pval(-s_null, -a_obs)))
```

```
##     observed    null_mean p_two_sided
##      557.000      541.267        0.224
```

Within departments, 557 women admitted is close to the 541 expected.

The Cochran-Mantel-Haenszel test

The within-stratum null has mean $\sum_k E_k$ and variance $\sum_k V_k$, set by each department's margins. CMH compares

$$\frac{(\sum_k A_{11k} - \sum_k E_k)^2}{\sum_k V_k},$$

with χ^2_1. It is a function of T, so the stratified permutation test is its exact version.

```
mantelhaen.test(UCBAdmissions, correct = FALSE)$p.value
```

```
## [1] 0.2169
```

Pooled, the data reject independence. Within departments, they do not.

Ordered categories and scores

Give each ordered category a **score** s, and use $T = \sum_i s_i Z_i$.

If a hidden Y_i orders everyone and each category is a range of Y, the best guess for a rank is the middle of the category's ranks, the **midrank**:

$$\{\#\text{ in lower categories}\} + \frac{\{\#\text{ in this category}\} + 1}{2}.$$

These are deck 4's ranks with ties, and T is a rank-sum statistic.

Example: alcohol and esophageal cancer

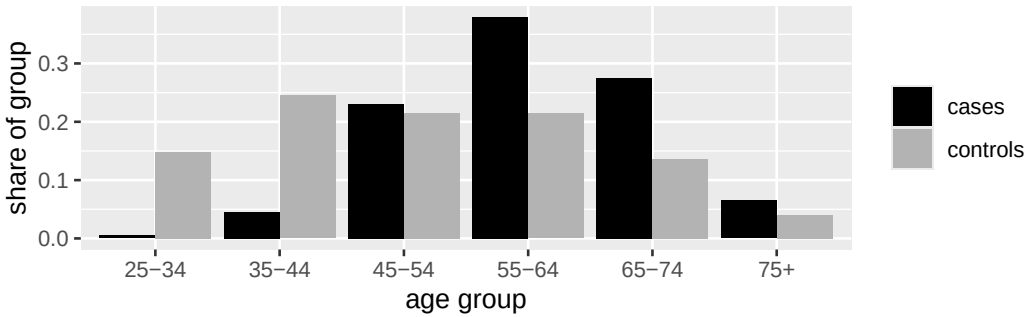
`esoph` (shipped with R): a case-control study in France, 200 cases and 775 controls.

```
eso <- bind_rows(esoph %>% uncount(ncases) %>% mutate(case = TRUE),
                 esoph %>% uncount(ncontrols) %>% mutate(case = FALSE))
with(eso, table(alcohol = alcgp, case))
```

```
##             case
## alcohol      FALSE TRUE
##   0-39g/day    386   29
##   40-79        280   75
##   80-119        87   51
##   120+          22   45
```

Half of the controls drink less than 40 g a day, against 14% of the cases.

Ages of cases and controls



Cases are older, and drinking varies with age, so compare within age groups.

H_0: within each age group, cases and controls have the same distribution of alcohol use.

Midrank scores for alcohol groups

```
eso <- eso %>% mutate(s = rank(as.integer(alcgp)))     # midranks
eso %>% count(alcgp, s) %>%
  mutate(lower = cumsum(n) - n, from_n = lower + (n + 1) / 2)
```

```
##       alcgp     s   n lower from_n
## 1 0-39g/day 208.0 415     0  208.0
## 2     40-79 593.0 355   415  593.0
## 3    80-119 839.5 138   770  839.5
## 4      120+ 942.0  67   908  942.0
```

`from_n`, the formula, matches `rank()`. Everyone in a category gets the same score.

$T = \sum_i s_i Z_i$ with $Z_i = 1$ for a case: the sum of the cases' midranks.

Rank-sum test within age groups

```
t_obs <- with(eso, sum(s * case))
nulls <- with(eso, cbind(pooled = replicate(B, sum(s * sample(case))),
  within_age = replicate(B, sum(s * shuffle_within(case, agegp)))))
data.frame(mean = colMeans(nulls), sd = apply(nulls, 2, sd),
           p = apply(nulls, 2, pval, t_obs = t_obs))
```

```
##                mean   sd     p
## pooled       97640 3392 0.001
## within_age 100811 3223 0.001
```

Observed $T = 135{,}712$, 10.8 SDs above the within-age null mean. $p = 0.001$ is the smallest possible with $B = 999$.

Summary

Summary: pairwise statistics

Test	A_{ij}	B_{ij}
Energy	distance	label weights
Nearest neighbor	j is i's nearest neighbor	same label
Two networks (QAP)	tie in network 1	tie in network 2
Node attribute	tie	attribute gap
Distance covariance	centered distance in X	centered distance in Y

$T = \sum_{i,j} A_{ij} B_{ij}$. Relabel rows and columns of B together. Compute distances, neighbors and centering once.

Summary: rearrangements

- **Equal distributions**: shuffle labels. Use all coordinates at once, and choose the units.
- **Independence**: shuffle the pairing. r measures only linear dependence; dCor is 0 only under independence.
- **Networks**: relabel the nodes, rows and columns together.
- **Strata**: shuffle within strata, the CMH test.
- **Ordered categories**: midrank scores.